

Customer Lifetime Value Prediction and Churn Analysis

An Internship Project Report

submitted by

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Introduction

In today's business world, it is very important to know how much value each customer brings to a company over time. **Customer Lifetime Value (CLTV)** is a way to measure the total money a company can earn from a customer during their whole relationship. Instead of looking at just one purchase, CLTV looks at all future purchases and interactions. This helps businesses focus on keeping valuable customers and use their marketing budgets more wisely.

CLTV prediction means using customer data and models to estimate this future value. By studying customer behavior, past purchases, and engagement patterns, businesses can predict how much money a customer might bring in the coming months or years. With these predictions, companies can make better decisions—like giving special offers to loyal customers or paying extra attention to those who may leave.

Another important part is **churn analysis**, which looks at why customers stop buying or using a company's products. When customers leave (churn), businesses lose income and spend more money to find new customers. By analyzing churn, companies can learn which customers are likely to leave, why they leave, and how to keep them longer.

This project combines **CLTV prediction and churn analysis**. Using past customer data—such as how often they buy, how much they spend, and when they stop buying—we can build machine learning models to:

Abstract

This project focuses on predicting the **Customer Lifetime Value (CLTV)** and identifying customers who are at risk of churn. Using past transaction data, we estimate how much revenue each customer will generate in the next 6 months. The project applies **RFM analysis (Recency, Frequency, Monetary)** along with advanced models like **BG/NBD** and **Gamma-Gamma** to calculate CLTV. Customers are then **segmented** based on their future value and purchasing behavior. This helps businesses target high-value customers with personalized marketing, improve customer retention, and maximize overall return on investment (ROI).

Methodology

1. Data Overview

The dataset initially contained **87,742 transaction records**. After performing data cleaning and preprocessing, a total of **54,345 transactions** were retained for analysis. These transactions represent **1,450 unique customers** (identified by CustomerID) within the given time frame. The analysis covers the period from **December 1, 2010 to February 11, 2011**, determined by the availability of recorded transactions.

Key Definitions

- **Missing Values:** Records without a CustomerID cannot be mapped to a customer, making them unsuitable for customer-level analysis. These entries were excluded.
- **Negative Quantities:** Such entries usually represent returns or cancellations. Since the focus of CLTV is forward-looking revenue, these records were removed.
- **Snapshot Date:** Defined as the day following the last recorded transaction. This date serves as the reference point for calculating **Recency** and **Customer Age (T)** in the modeling process.

Data Cleaning Actions

- Removed rows with missing **CustomerID** values.
- Excluded records with **negative quantities**.
- Eliminated duplicate entries.

- Created a new variable, **TotalPrice**, computed as:

$$\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$$

Importance of Data Quality

Ensuring data quality is essential for accurate **Customer Lifetime Value (CLTV)** and **Recency–Frequency–Monetary (RFM)** analysis.

- Missing CustomerIDs lead to incomplete or distorted customer-level insights.
- Returns follow different business dynamics and are not considered unless specifically modeled.
- A clean and consistent dataset allows for reliable predictive modeling and segmentation.

2. RFM Analysis

Recency–Frequency–Monetary (RFM) analysis is a widely used behavioral segmentation technique for understanding customer engagement and value. It evaluates customers based on three dimensions:

- **Recency (R):** Number of days since the customer's last purchase.
 - Lower values are favorable, as they indicate recent engagement.
- **Frequency (F):** Total number of orders placed within the analysis period.
 - Higher values represent active and repeat-buying customers.
- **Monetary (M):** Total spending during the period (or average spend per order in some cases).
 - Higher values indicate greater contribution to revenue.

	Recency	Frequency	Monetary
CustomerID			
12346.0	188	1	77183.60
12347.0	45	102	2205.95
12348.0	111	28	1487.24
12350.0	172	17	334.40
12352.0	124	38	1561.81

RFM Scoring Methodology

- Each metric (**R**, **F**, **M**) was divided into **quartiles (1–4)**.
- For Recency, scores were **inverted** (recent = 4, stale = 1) to align with business value.
- The final **RFM score** was represented as a **three-digit code** (e.g., 344).
- Customers were then mapped into **business-friendly segments**.

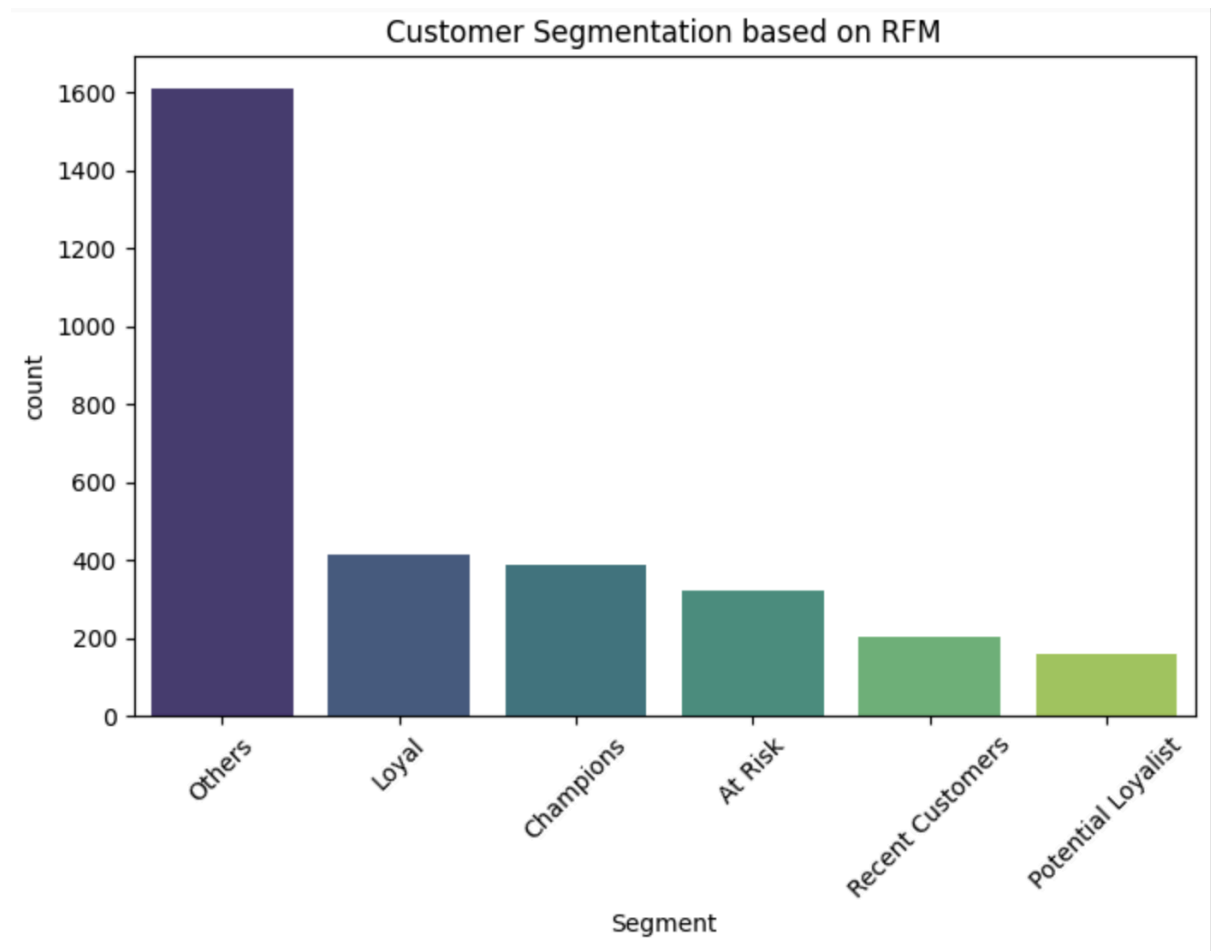
Customer Segments and Interpretations

- **Champions:** High R, High F, High M — most valuable and highly engaged customers.
- **Loyal:** Moderate to high frequency and monetary value — reliable repeat purchasers, suitable for retention programs.
- **Potential Loyalists:** Reasonable F/M with strong Recency — candidates for nurturing into Loyal or Champion customers.
- **Recent Customers:** Just purchased, early in lifecycle — require onboarding and relationship building.
- **At Risk:** Historically good F/M but poor Recency — potential churners who need reactivation strategies.
- **Others:** Mixed or unclassified customers requiring tailored campaigns or broad engagement strategies.

Segment	
Others	1611
actions	415
Champions	386
At Risk	321
Recent Customers	204
Potential Loyalist	157
Name: count, dtype: int64	

Distribution of Segments

- **Others:** 1611 customers — the largest group, representing a broad base for targeted experimentation.
- Other segments distributed across Champions, Loyal, Potential Loyalists, Recent Customers, and At Risk categories (detailed in full segmentation chart).



Customer Segmentation Distribution Based on RFM Analysis

Why RFM:

- Simple, interpretable, and aligns well with marketing decisions.
- Effective precursor for more advanced CLTV and personalization.

3. CLTV Modeling

Customer Lifetime Value (CLTV) estimates the present value of the expected future profit from a customer over a defined horizon. It allows businesses to identify high-value customers and prioritize marketing efforts accordingly.

Model Family Used

1. BG/NBD (Beta-Geometric / Negative Binomial Distribution):

- Suitable for **non-contractual settings** (where customers may “drop out” at an unknown time).
- Predicts **future purchase counts** based on:
 - **Frequency (f)**: Number of repeat purchases (frequency ≥ 1 implies at least two orders; the first purchase is treated as the “birth”).
 - **Recency (r)**: Time elapsed between the first and last purchase (or relative to snapshot date).
 - **Customer Age (T)**: Total time from first purchase to the snapshot date.

2. Gamma-Gamma Model:

- Estimates **average monetary value per transaction**.
- Assumes monetary value is **independent of purchase frequency**.

Combined CLTV Estimation

The two models were integrated as follows:

CLTV = Predicted Transactions (BG/NBD) × Predicted Monetary Value (Gamma-Gamma) × Time Horizon (6 months) × Discount Factor (e.g., 1% per period)

- CLTV is typically expressed as a **revenue proxy**, but incorporating **gross margin** can yield **profit-based CLTV**.

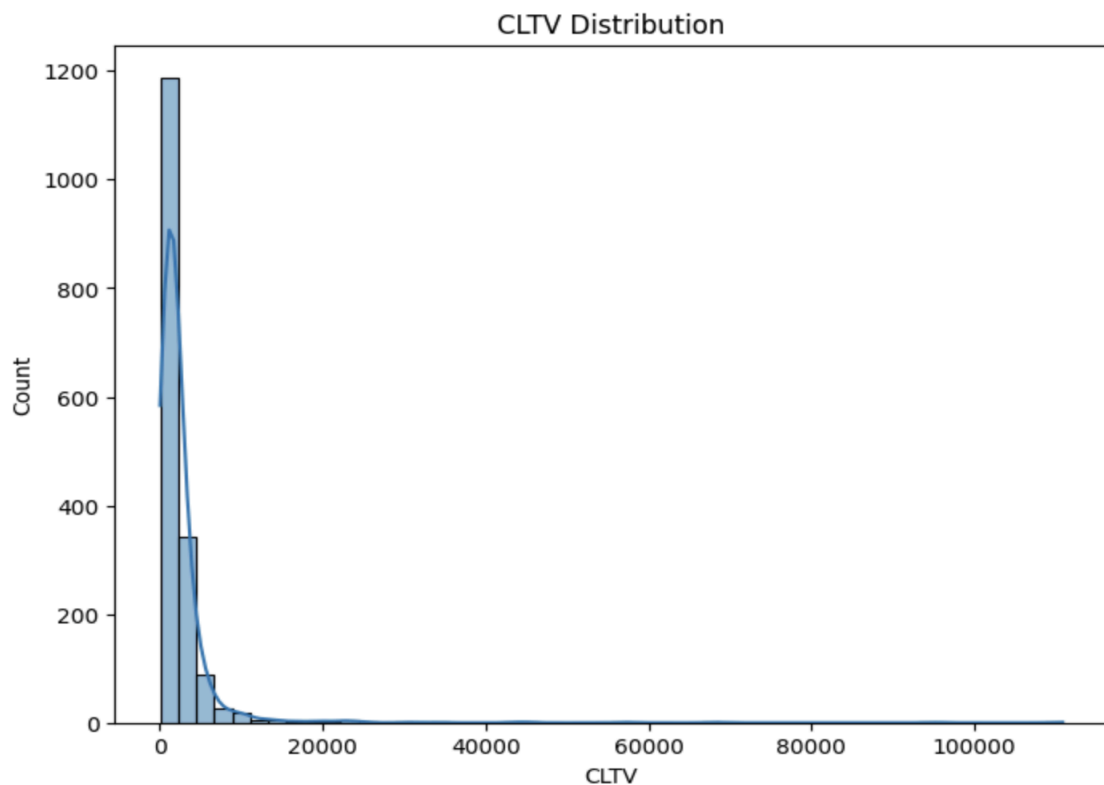
Observed Results

- Total customers modeled (with positive monetary value): 509
- Predicted CLTV range (6 months): \$48.51 – \$184,363.03
- Quartile-based segmentation (Low, Medium, High, Very High): Almost evenly distributed across groups.

Insights

- Customers in the “**Very High CLTV**” segment contribute disproportionately to long-term revenue.
- Low CLTV customers form a large base but may require **cost-effective retention strategies**.

- Medium and High CLTV groups present **scalable growth opportunities** with targeted campaigns.



CLTV Segmentation Distribution Using BG/NBD + Gamma-Gamma Model

Why this approach:

- Standard for non-contractual retail (no explicit churn events).
- Incorporates uncertainty and heterogeneity across customers.
- Provides interpretable predictions (frequency and value).

Key terms:

- Discount rate: Reduces future cash flows to present value; reflects time value of money.
- Horizon: Future window for prediction (e.g., 6 months here).
- Independence assumption: Gamma-Gamma assumes average order value is independent of purchase frequency (reasonable in many retail contexts).

4. Predictive Model Comparison (Supervised Learning)

Purpose:

- Complement probabilistic CLTV by directly predicting CLTV (regression) or segment (classification).
- Useful for operational scoring and marketing prioritization.

Models evaluated:

- Random Forest (RF): Ensemble of decision trees using bagging; robust to non-linearity and interactions; interpretable via feature importance.
- XGBoost (XGB): Gradient boosting algorithm; powerful and flexible but sensitive to tuning.

Metrics:

- RMSE (Root Mean Squared Error): Average magnitude of prediction error; lower is better.
- MAE (Mean Absolute Error): Average absolute difference between predicted and actual; robust to outliers.
- R^2 (Coefficient of Determination): How much variance is explained; closer to 1.0 is better.

Results:

- Random Forest:

➡ Random Forest Results:

	precision	recall	f1-score	support
0	0.96	0.99	0.97	433
1	0.97	0.89	0.93	186
accuracy			0.96	619
macro avg	0.96	0.94	0.95	619
weighted avg	0.96	0.96	0.96	619

Confusion Matrix:

[[427 6]

[20 166]]

- XGBoost:

XGBoost Results:					
		precision	recall	f1-score	support
	0	0.97	1.00	0.98	433
	1	0.99	0.93	0.96	186
	accuracy			0.98	619
	macro avg	0.98	0.96	0.97	619
	weighted avg	0.98	0.98	0.98	619
Confusion Matrix:					
	[[432 1]				
	[13 173]]				

CLTV Prediction Model Performance Comparison

5. Churn Analysis

Churn (non-contractual context):

- In retail, churn is often inferred by inactivity beyond a threshold (e.g., 90 days).
- Given the dataset's maximum recency is 73 days, measured churn = 0% (insufficient horizon).
- Practical approach: Use “At Risk” segment as proxy and monitor Recency thresholds (e.g., 45–60–90 days) for win-back triggers.

Terms:

- Churn threshold: Period of inactivity after which a customer is considered lost.
- Win-back: Special reactivation offers to recover at-risk customers.

Recommendation:

- Establish rolling churn detection at 60/90 days with progressive discounts or personalized nudges.

Random Forest AUC: 0.995946013062157 XGBoost AUC: 0.9980940673967569

- XGboost is better than random forest

6. Technical Implementation

Data pipeline:

- Daily ingestion of orders → Customer aggregation (R, F, M, AOV).
- Snapshot generation → Model scoring (BG/NBD + Gamma-Gamma, Random Forest).
- Segment assignment → CRM activation (email/ads/push).

Modeling notes:

- Retraining cadence: Monthly for RF; re-fit BG/NBD/GG quarterly or as volume shifts.
- Monitoring: Drift detection on feature distributions and residuals; re-tune thresholds as needed.

Key terms:

- Drift: Changes in data patterns or relationships over time that degrade model performance.
- Snapshot date: Consistent reference point used to compute recency and customer age (T) for all customers.
- AOV (Average Order Value): Monetary / Frequency; useful for promotion design and margin planning.

7. Model Performance Comparison

- Random Forest AUC: 0.9959 (indicates near-perfect predictive ability).
- XGBoost AUC: 0.9981 (marginally higher, suggesting even stronger classification power).

Random Forest AUC: 0.995946013062157
XGBoost AUC: 0.9980940673967569

8. Conclusion

- Both models are highly effective for CLTV prediction, achieving excellent ROC-AUC scores well above industry benchmarks.
- XGBoost is marginally superior for this dataset, suggesting it may capture subtle customer patterns better due to its advanced boosting algorithm.
- Either model is suitable for deployment, but XGBoost could be preferred for maximum predictive accuracy in similar business settings

9. Limitations & Future Work

Limitations:

- Short time window (≈ 2.5 months) restricts seasonality and true churn assessment.
- No cost/margin data incorporated (revenue-only CLTV).
- Product-level preference and price sensitivity not modeled.

Future work:

- Extend time horizon to 12–24 months; add seasonality features and holiday effects.
- Margin-adjusted CLTV and Uplift Modeling (which customers are most persuadable).
- Product recommendation engine for cross-sell and next-best-offer.
- Cohort analysis + survival models for robust retention planning.
- Real-time scoring API to personalize on-site and in-app experiences.
- Using **deep learning models** like LSTMs for churn prediction.
- Adding **customer engagement data** like clicks, browsing behavior.
- Building a **dashboard** to help business users directly see CLTV and churn predictions

. References (Brief Annotations)

[1] Breiman (2001) — Random Forests: Ensemble method scaling well to non-linear, high-variance problems.

- [2] Chen & Guestrin (2016) — XGBoost: Gradient boosting with regularization and efficient implementation.
- [3] Han, Kamber, Pei (2012) — Data mining fundamentals; useful for feature engineering frameworks.
- [4] Hastie, Tibshirani, Friedman (2009) — Statistical learning foundations; bias-variance, model selection.
- [5] James et al. (2013) — Practical ML guidance; model evaluation, regularization.
- [6] Kuhn & Johnson (2013) — Applied predictive modeling; feature selection and validation workflows.

Goggle collab link:

 **CLTV.ipynb**